

Biconnected Graph Triangulation

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Abstract

First we will show the higher level pseudocode of our algorithm for enumerating all possible triangulations of a biconnected plane graph. Then we will illustrate the steps of the algorithm with an example graph.

FindAllTriangulations

Algorithm FindAllTriangulations

- 1: **procedure** FINDALLTRIANGULATIONS(G, k)
 - 2: Label faces as $F_1, F_2, \dots, F_k$ arbitrarily
 - 3: Initialize global chord set S
 - 4: $T \leftarrow \emptyset$ ▷ Empty partial triangulation
 - 5: FINDALLTRIANGULATIONS_CYCLE($F, T, 0$)
 - 6: **end procedure**
-

FindSafeRootTriangulation

Algorithm FindSafeRootTriangulation

```
1: procedure FINDSAFEROOTTRIANGULATION( $F$ , index)
2:   startIndex  $\leftarrow n - 1$ 
3:   endIndex  $\leftarrow 0$ 
4:   while startIndex > endIndex + 1 do
5:     if chord ( $v_{\text{startIndex}}, v_{\text{endIndex}}$ ) exists in  $S$  then
6:       startIndex  $\leftarrow$  startIndex - 1
7:     else
8:       endIndex  $\leftarrow$  endIndex + 1
9:     end if
10:  end while
11:   $v_r \leftarrow v_{\text{endIndex}}$  ▷ Root vertex for safe triangulation
12:  Build fan triangulation  $T_r$  from  $v_r$  as root vertex
13:  return  $T_r$ 
14: end procedure
```

FindAllTriangulationsCycle

Algorithm FindAllTriangulationsCycle

```
1: procedure FINDALLTRIANGULATIONS-cycle( $F, T, i$ )
2:    $T_i \leftarrow \text{FINDSAFEROOTTRIANGULATION}(F, i)$   $\triangleright \mathcal{O}(n)$ 
3:   if  $F_i$  is the last face then  $\triangleright$  All faces triangulated
4:     output ( $T \cup T_i$ )
5:   else
6:     FINDALLTRIANGULATIONS-cycle( $F, T \cup T_i, i + 1$ )
7:   end if
8:   for  $j = n - 1$  down to 3 do
9:     FINDALLCHILDTRIANGULATIONS-cycle( $T_i(v_r, v_j), T, i$ )
10:  end for
11: end procedure
```

FindAllChildTriangulationsCycle

Algorithm FindAllChildTriangulationsCycle

```
1: procedure FINDALLCHILDTRIANGULATIONS-cycle( $T_i, T, i$ )
2:   if  $F_i$  is the last face then                                ▷ All faces triangulated
3:     output ( $T \cup T_i$ )
4:   else
5:     FINDALLTRIANGULATIONS-cycle( $F, T \cup T_i, i + 1$ )
6:   end if
7:   Let  $(v_b, v_{b'})$  be the leftmost generating chord of  $T_i$ 
8:   for  $j$  from  $b$  to  $n - 1$  do
9:     if  $(v_r, v_j)$  is a chord of  $T_i$  and  $\text{oppositePair}(v_r, v_j) \notin S$  then
10:      FINDALLCHILDTRIANGULATIONS-cycle( $T_i(v_r, v_j), T, i$ )
11:    end if
12:  end for
13: end procedure
```

A biconnected plane graph with three faces

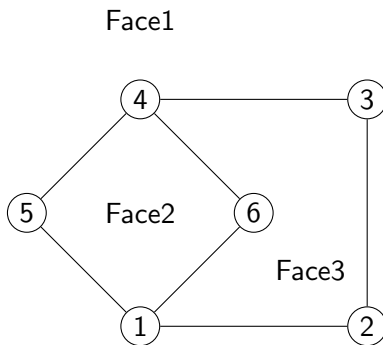
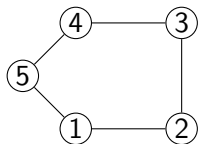
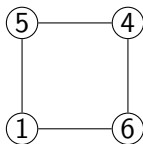


Figure: Root triangulation with safe root vertex 1. Chords from vertex 1 to all non-neighboring vertices (3, 4, 5) form a fan structure, ensuring no multi-edge conflicts with previously added chords.

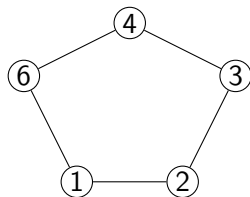
Faces of the Graph



(a) Face 1

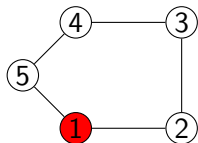


(b) Face 2

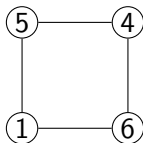


(c) Face 3

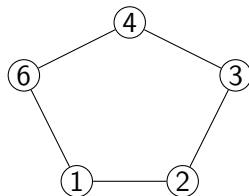
Find safe root of face 1



(a) **Face 1**



(b) Face 2

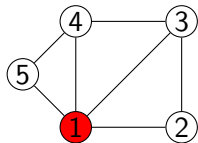


(c) Face 3

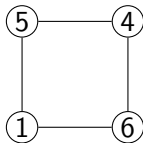
Global Edge Set:

$(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6)$

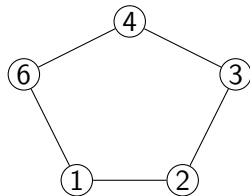
Root Triangulation of face 1



(a) **Face 1**



(b) Face 2

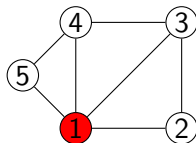


(c) Face 3

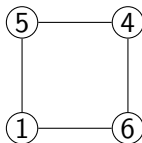
Global Edge Set:

$(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), (1,3), (1,4)$

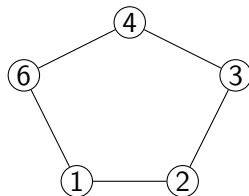
Now we move to face 2



(a) Face 1



(b) **Face 2**

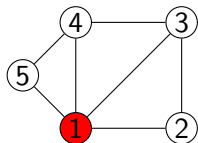


(c) Face 3

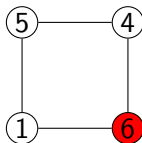
Global Edge Set:

$(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), (1,3), (1,4)$

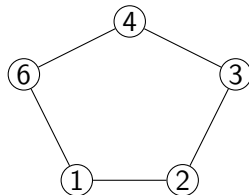
Find safe root of face 2



(a) Face 1



(b) **Face 2**

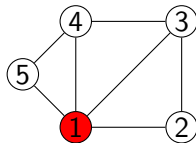


(c) Face 3

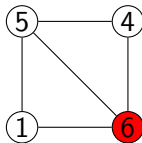
Global Edge Set:

(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(1,3), (1,4)**

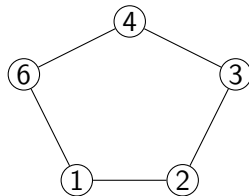
Root Triangulation of face 2



(a) Face 1



(b) **Face 2**

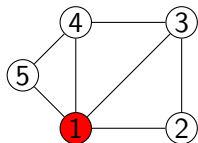


(c) Face 3

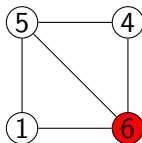
Global Edge Set:

(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(1,3), (1,4), (5,6)**

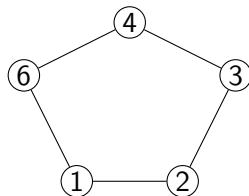
Now we move to face 3



(a) Face 1



(b) Face 2

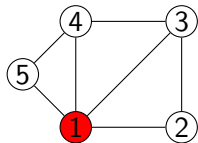


(c) **Face 3**

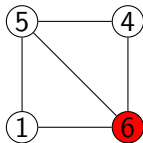
Global Edge Set:

(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(1,3), (1,4), (5,6)**

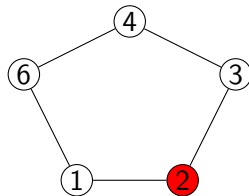
Find safe root of face 3



(a) Face 1



(b) Face 2

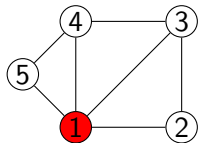


(c) **Face 3**

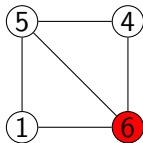
Global Edge Set:

(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(1,3), (1,4), (5,6)**

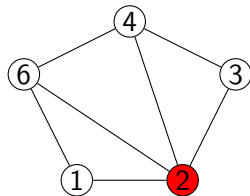
Root Triangulation of face 3



(a) Face 1



(b) Face 2

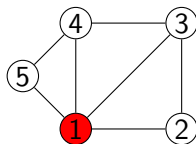


(c) **Face 3**

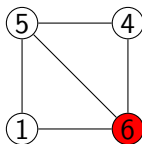
Global Edge Set:

(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), (1,3), (1,4), (5,6). (2,4), (2,6)

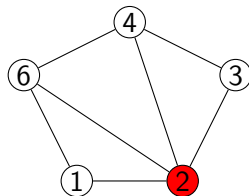
There is no other faces after face 3, so its a complete valid triangulation



(a) Face 1



(b) Face 2

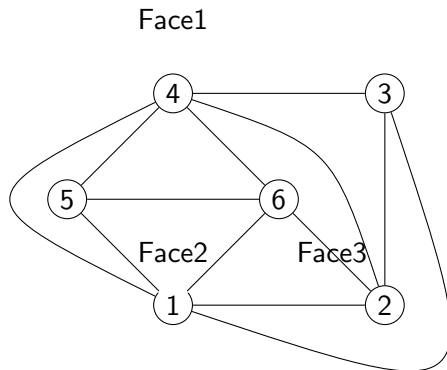


(c) Face 3

Global Edge Set:

$(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), (1,3), (1,4), (5,6), (2,4), (2,6)$

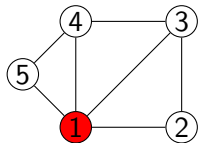
So the actual triangulation looks like this:



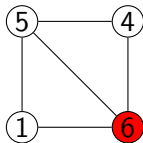
Global Edge Set:

(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), (1,3), (1,4), (5,6), (2,4), (2,6)

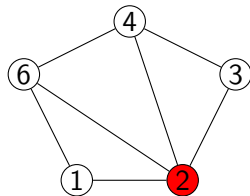
Now we will start flipping of Face 3



(a) Face 1



(b) Face 2

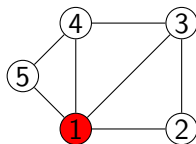


(c) Face 3

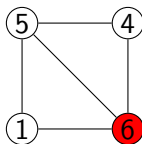
Global Edge Set:

(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), (1,3), (1,4), (5,6). (2,4), (2,6)

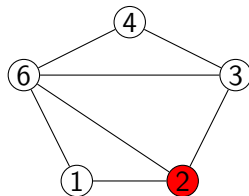
We can flip the (2,4) edge, as its opposite pair is not in the global edge set



(a) Face 1



(b) Face 2

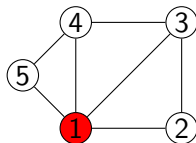


(c) Face 3

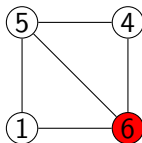
Global Edge Set:

(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), (1,3), (1,4), (5,6). (3,6), (2,6)

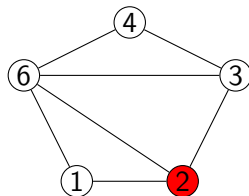
Its a new triangulation



(a) Face 1



(b) Face 2

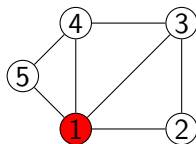


(c) **Face 3**

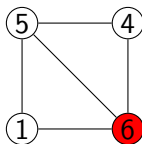
Global Edge Set:

(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), (1,3), (1,4), (5,6). (3,6), (2,6)

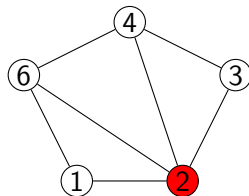
in the root also, cant flip (2,6) as (1,4) is present in the global edge set



(a) Face 1



(b) Face 2

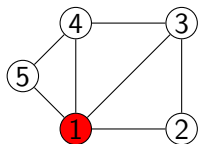


(c) Face 3

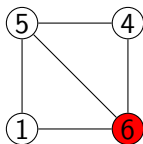
Global Edge Set:

(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), (1,3), (1,4), (5,6). (2,4), (2,6)

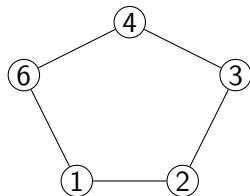
So, there is no more triangulations possible from here, so we backtrack to face 2



(a) Face 1



(b) Face 2

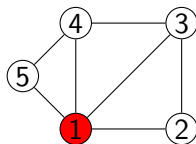


(c) Face 3

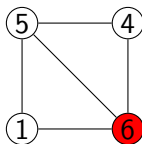
Global Edge Set:

$(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), (1,3), (1,4), (5,6)$

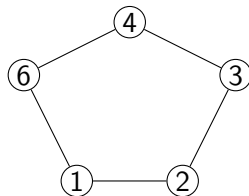
We can't flip the (5,6) edge as (1,4) is present in the global edge set



(a) Face 1



(b) Face 2

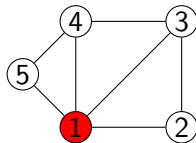


(c) Face 3

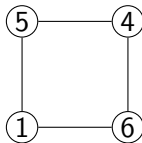
Global Edge Set:

(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), (1,3), (1,4), (5,6)

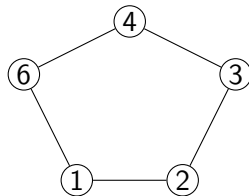
So we backtrack to face 1



(a) **Face 1**



(b) Face 2

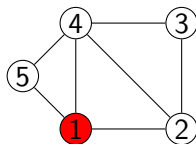


(c) Face 3

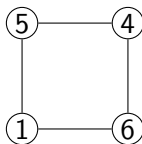
Global Edge Set:

$(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), (1,3), (1,4)$

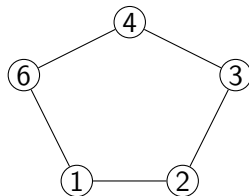
We can flip the (1,3) edge, as its opposite pair is not in the global edge set



(a) **Face 1**



(b) Face 2

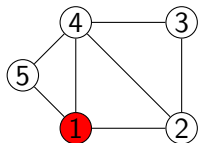


(c) Face 3

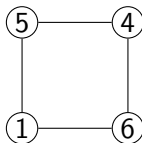
Global Edge Set:

(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(2,4)**, **(1,4)**

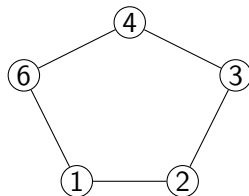
Now we move to face 2



(a) Face 1



(b) **Face 2**

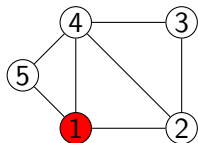


(c) Face 3

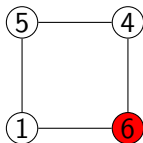
Global Edge Set:

$(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), (2,4), (1,4)$

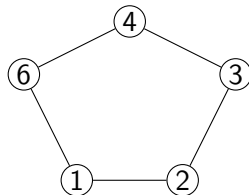
Find the safe root of face 2



(a) Face 1



(b) **Face 2**

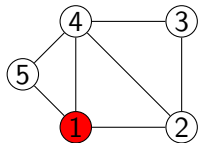


(c) Face 3

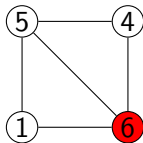
Global Edge Set:

(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(2,4)**, **(1,4)**

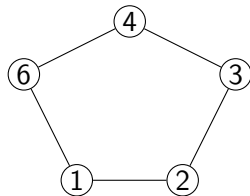
Root Triangulation of face 2



(a) Face 1



(b) **Face 2**

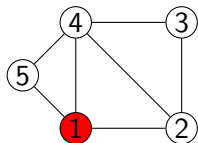


(c) Face 3

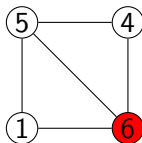
Global Edge Set:

(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(2,4)**, **(1,4)**, **(5,6)**

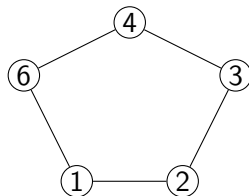
Now we move to face 3



(a) Face 1



(b) Face 2

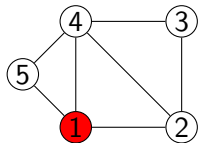


(c) **Face 3**

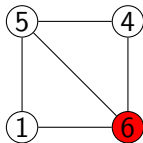
Global Edge Set:

(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), (2,4), (1,4), (5,6)

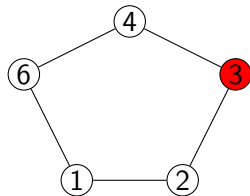
Find safe root of face 3



(a) Face 1



(b) Face 2

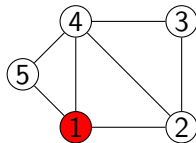


(c) **Face 3**

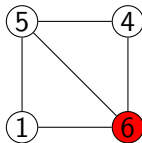
Global Edge Set:

(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), (2,4), (1,4), (5,6)

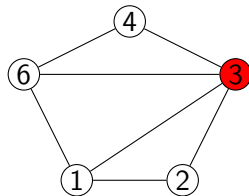
Root Triangulation of face 3



(a) Face 1



(b) Face 2

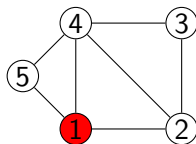


(c) Face 3

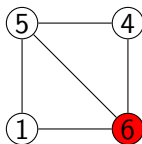
Global Edge Set:

(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), (2, 4), (1,4), (5,6), (1,3), (3,6)

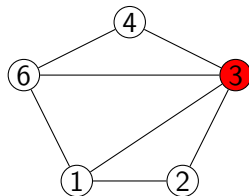
There is no face after face 3, so its a complete triangulation



(a) Face 1



(b) Face 2

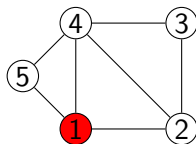


(c) Face 3

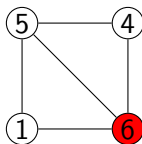
Global Edge Set:

(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), (2, 4), (1,4), (5,6), (1,3), (3,6)

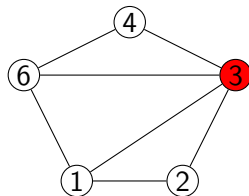
Cant flip (3,6), because opposite pair is (1,4), already exists



(a) Face 1



(b) Face 2

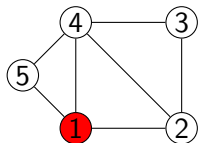


(c) Face 3

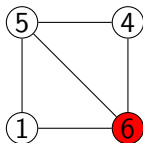
Global Edge Set:

(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), (2, 4), (1,4), (5,6), (1,3), (3,6)

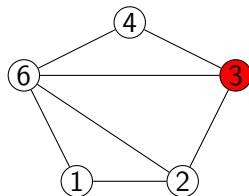
we can flip (1,3) to (2,6), its a new triangulation



(a) Face 1



(b) Face 2

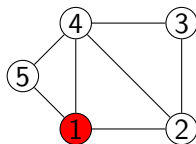


(c) Face 3

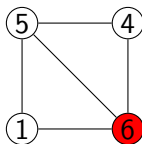
Global Edge Set:

(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), (2, 4), (1,4), (5,6), (2,6), (3,6)

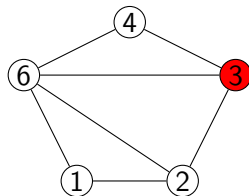
Cant flip (3,6), because opposite pair is (2,4), already exists



(a) Face 1



(b) Face 2

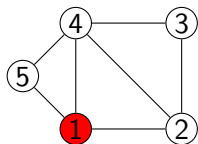


(c) Face 3

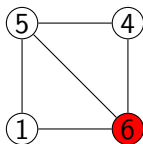
Global Edge Set:

(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), (2, 4), (1,4), (5,6), (2,6), (3,6)

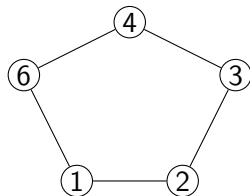
So, there is no more triangulations possible from here, so we backtrack to face 2



(a) Face 1



(b) Face 2

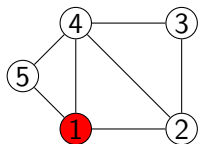


(c) Face 3

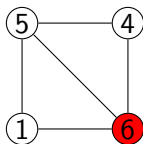
Global Edge Set:

$(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), (2,4), (1,4), (5,6)$

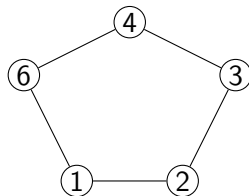
We can't flip the (5,6) edge as (1,4) is present in the global edge set



(a) Face 1



(b) Face 2

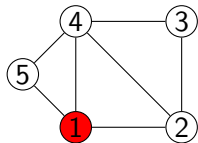


(c) Face 3

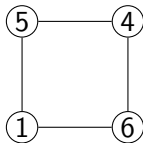
Global Edge Set:

(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), (2, 4), (1,4), (5,6)

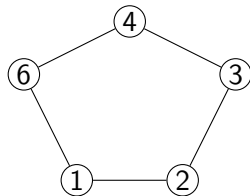
So we backtrack to face 1



(a) Face 1



(b) Face 2

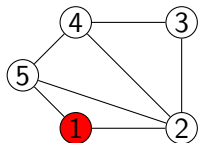


(c) Face 3

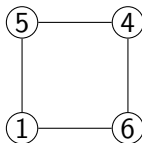
Global Edge Set:

$(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), (2,4), (1,4)$

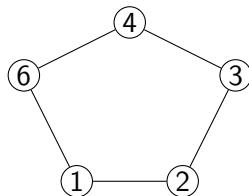
We can flip (1,4) to (2,5)



(a) Face 1



(b) Face 2

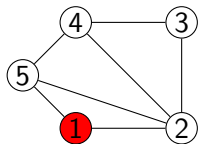


(c) Face 3

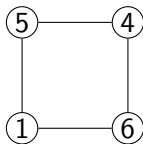
Global Edge Set:

(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(2,4)**, **(2,5)**

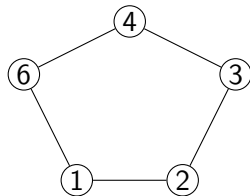
We can now move to Face 2



(a) Face 1



(b) **Face 2**

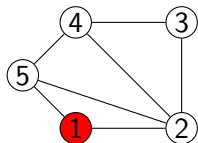


(c) Face 3

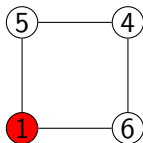
Global Edge Set:

$(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), (2,4), (2,5)$

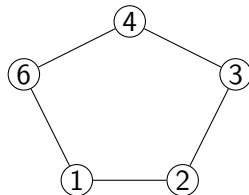
Find safe root of Face 2



(a) Face 1



(b) **Face 2**

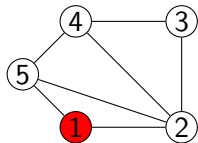


(c) Face 3

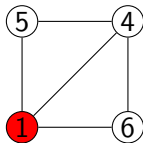
Global Edge Set:

$(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), (2,4), (2,5)$

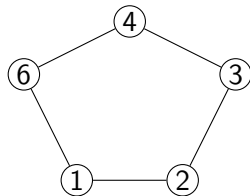
Root Triangulation of Face 2



(a) Face 1



(b) **Face 2**

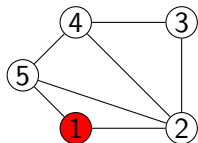


(c) Face 3

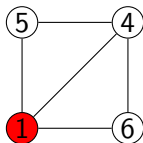
Global Edge Set:

(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), (2,4), (2,5), (1,4)

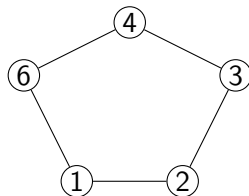
Now we move to Face 3



(a) Face 1



(b) Face 2

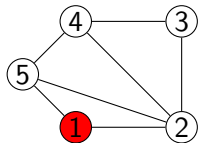


(c) **Face 3**

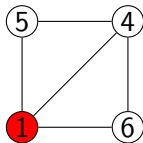
Global Edge Set:

(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), (2,4), (2,5), (1,4)

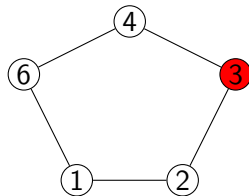
Find safe root of Face 3



(a) Face 1



(b) Face 2

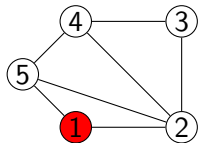


(c) **Face 3**

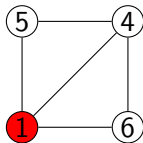
Global Edge Set:

(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), (2,4), (2,5), (1,4)

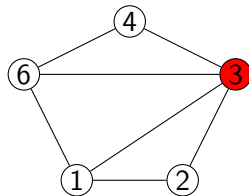
Root Triangulation of Face 3



(a) Face 1



(b) Face 2

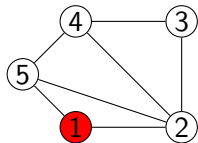


(c) **Face 3**

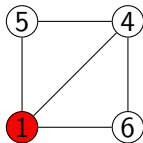
Global Edge Set:

(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), (2,4), (2,5), (1,4), (3,6), (1,3)

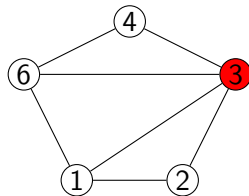
There is no more faces, so its a complete triangulation



(a) Face 1



(b) Face 2

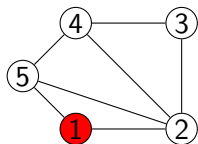


(c) Face 3

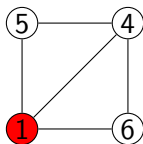
Global Edge Set:

$(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), (2,4), (2,5), (1,4), (3,6), (1,3)$

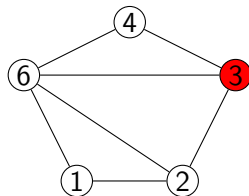
We can flip the (1,3) edge to (2,6) to get a different triangulation



(a) Face 1



(b) Face 2

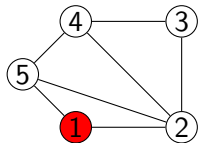


(c) Face 3

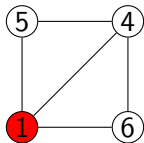
Global Edge Set:

(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), (2,4), (2,5), (1,4), (3,6), (2,6)

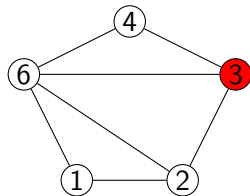
There is no more faces, so it's a complete triangulation



(a) Face 1



(b) Face 2



(c) Face 3

Global Edge Set:

$(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), (2,4), (2,5), (1,4), (3,6), (2,6)$

Continue....

We will be continuing this process until all possible triangulations are found.